uTile Demo PuTTY Manual

Revision 0

Date: 2016 October 3

This work is licensed under a Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 International License.

Table of Contents

1 Introduction.....3
1.1 PuTTY under Windows OS.....4
1.2 PuTTY under Linux OS.....6
2 Questions and Answers.....9

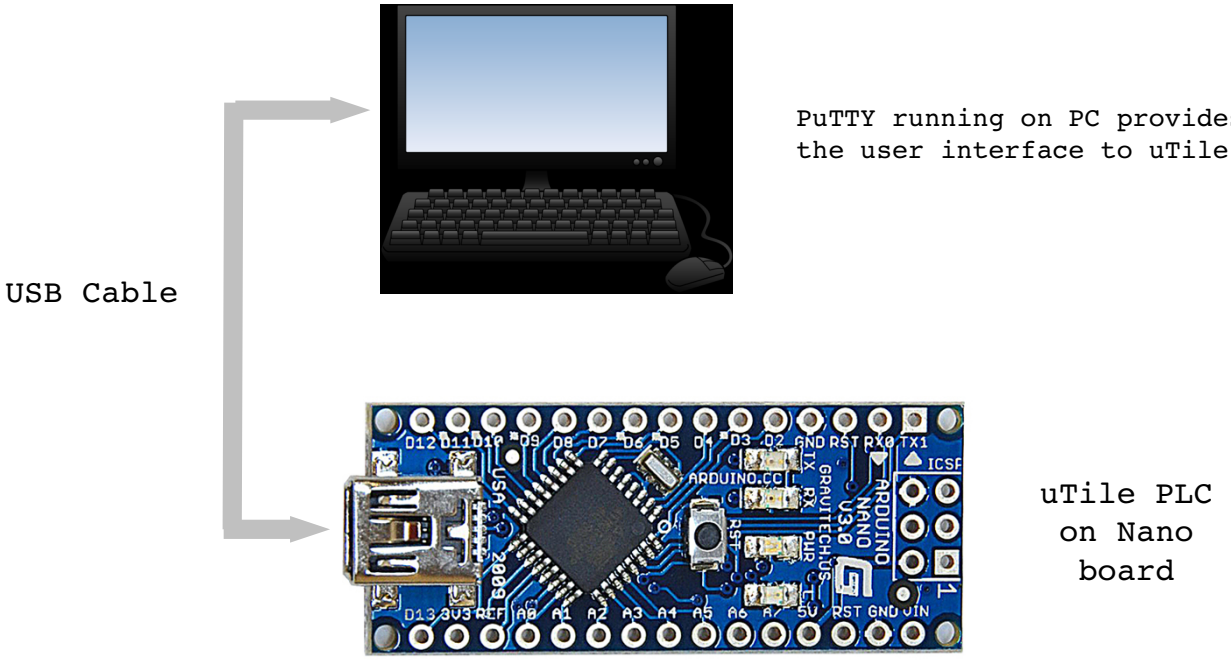
Copyright © 2016 - Francis Lyn

1 Introduction

This Manual describes how to install and run the PuTTY SSH and telnet client program as the human user interface (HUI) for the Demo version of the uTile Controller system. PuTTY is open-source software available for both Windows and *nix flavor operating systems. Since PuTTY runs on both of these popular operating systems, it is an ideal multi-platform HUI for use with the uTile PLC.

For Windows OS, PuTTY is available for download at #http://www.putty.org/. Linux distrubutions usually have PuTTY in the package repository so you can install the program using the package's installation tools.

Assuming you have installed PuTTY on your host PC and you are now ready to power-up and run the uTile PLC on the Arduino controller board, plug in a USB cable between the board and any USB connector on your host PC. Then follow the section below for the OS you are using. The sketch below shows a typical connection between PuTTY (on the host PC) and uTile on an Arduino Nano controller board. We are using the Nano board as an example in this manual.

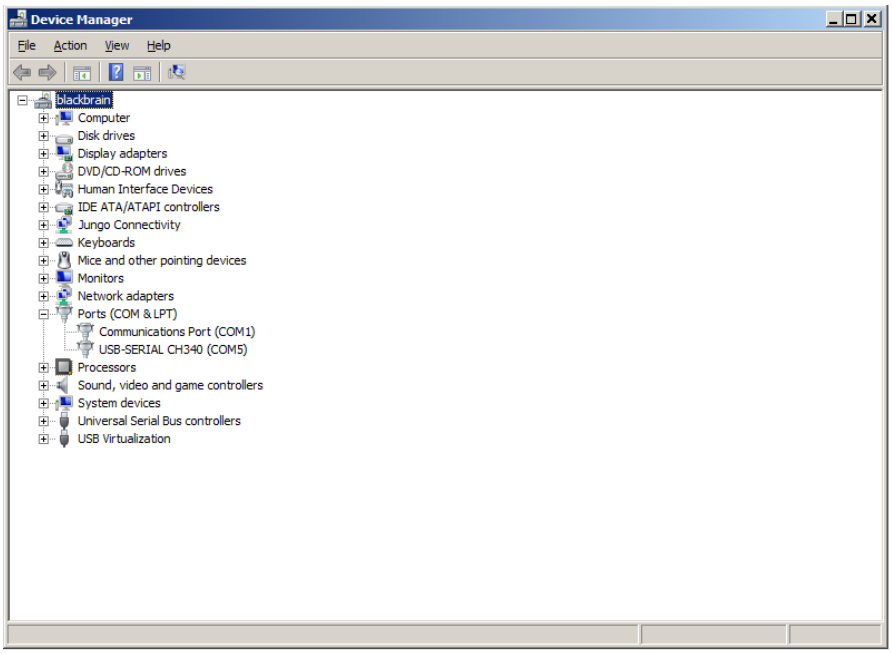


1.1 PuTTY under Windows OS

We will be using the terminal emulator function of PuTTY with a serial connection to uTile to provide the terminal interface. Connect the Arduino board to a PC USB port for the set-up procedure.

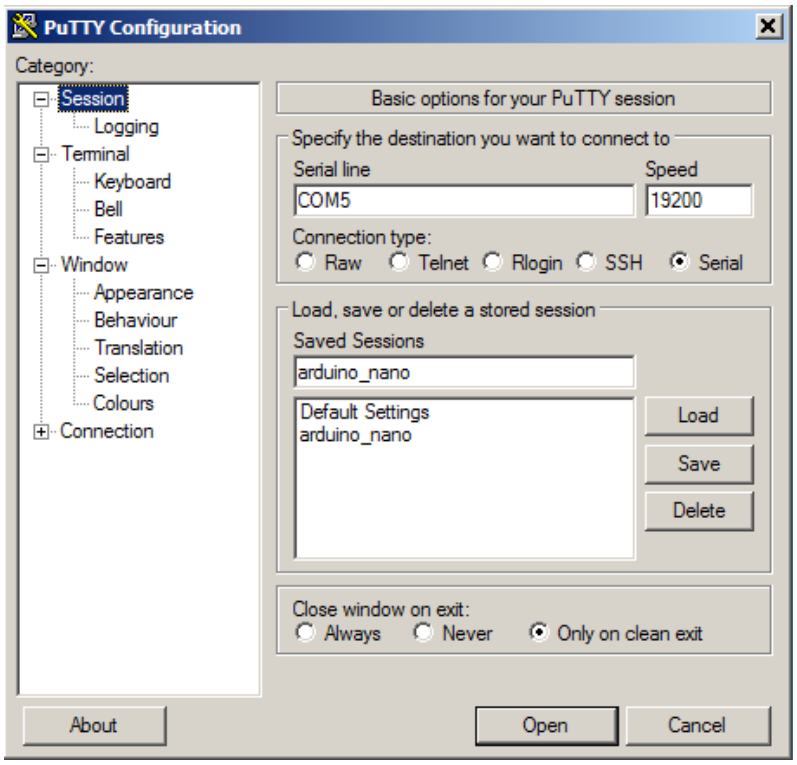
The first thing to do is to install the required USB device driver for the type of Arduino controller board you are running uTile on. You need to get and install the correct driver from the Arduino web site. Check other popular Arduino web sites like Lady Ada, Sparkfun, etc. for advice on setting up the Arduino board serial port driver.

You now need to determine which serial COM port the OS assigned to the plugged-in controller board. To find out which COM port is assigned to the plugged-in Arduino board, open the Windows Control Panel, select Hardware and Sound, and select Device Manager. You should see a screen somewhat like the following:



Under Ports (COM & LPT) folder you see USB-SERIAL CH340 (COM5) listed, which identifies COM5 as the assigned port for the Arduino Nano board. Your actual COM port may differ, depending on what other devices are plugged in to your USB ports at the time.

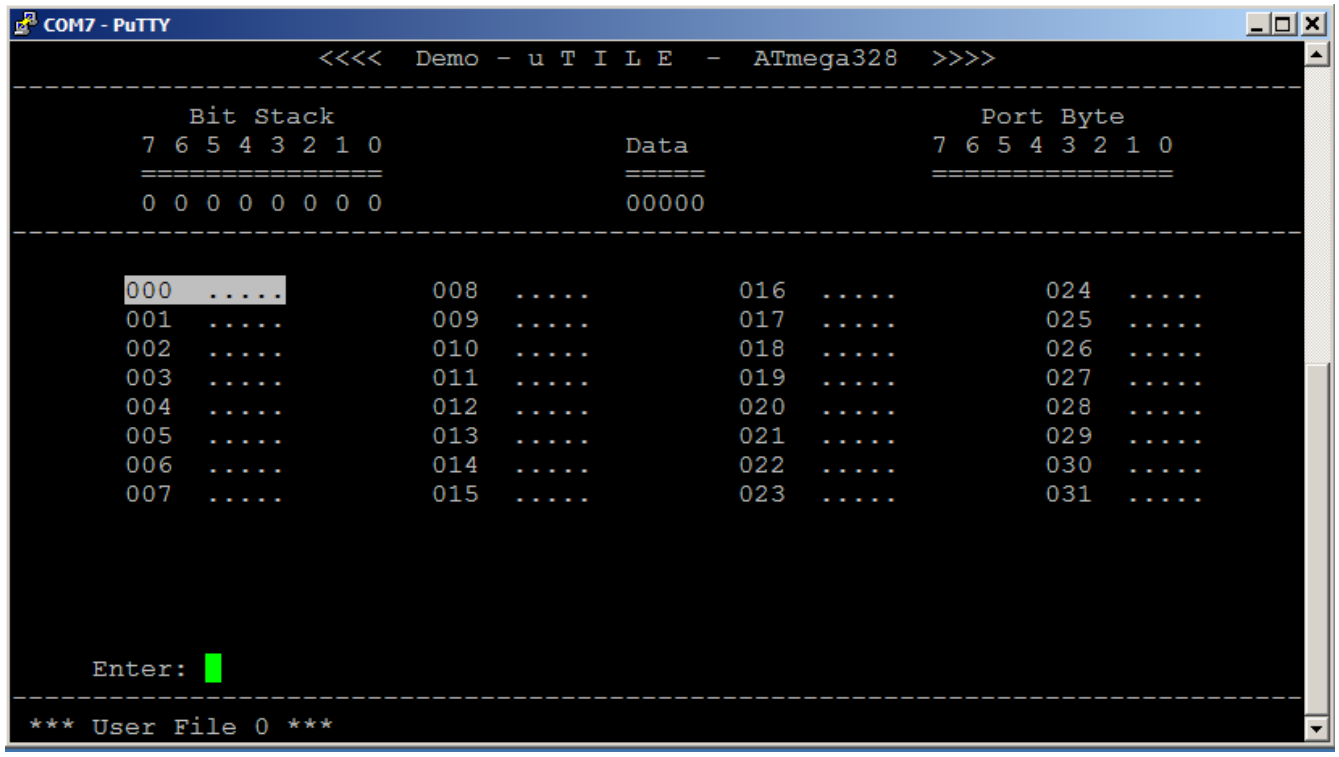
Start PuTTY and the configuration screen appears. Modify the settings as shown in the screen below:



Under the Category selection on the left pane, Connection, SSH, open the screen that shows the serial settings, and for the Flow control box select none. Go back to the Session screen and save the configuration used by providing a session name to use (arduino_nano used in this example) and clicking on the Save button.

This is the bare minimum required to get PuTTY to work with uTile. You can customize PuTTY further to suite your taste, like changing the terminal font size, screen size, etc. Save the session settings under a suitable name, such as arduino_nano so you can re-use the same settings the next time you start PuTTY.

The uTile main editor screen should appear when you select the Open tab and it should look like the screen below. Note the COM7 serial port on the top line of the PuTTY screen in this example.



1.2 PuTTY under Linux OS

To run PuTTY as a normal user it needs to be configured first. Connect the Arduino board to any PC USB port. We are assuming uTile on a Nano board and a Debian Linux distribution in the following description. Open a user terminal and type in the following command:

ls -l /dev/tty* | grep dialout

You should see a listing such as:

```
crw-rw---T 1 root dialout 166,  0 Sep 27 08:53 /dev/ttyUSB0
crw-rw---T 1 root dialout   4, 64 Sep 27 04:04 /dev/ttyS0
crw-rw---T 1 root dialout   4, 65 Sep 27 04:04 /dev/ttyS1
crw-rw---T 1 root dialout   4, 66 Sep 27 04:04 /dev/ttyS2
crw-rw---T 1 root dialout   4, 67 Sep 27 04:04 /dev/ttyS3
```

The first line in the listing shows ***/dev/ttyUSB0*** which is the serial

device assigned to the Arduino board serial connection. If you are using a different model Arduino, say an UNO or a Mega2560 you may see ***/dev/ttyACM0*** instead of ***/dev/ttyUSB0***. Make a note of this information as you will need to enter it into the PuTTY configuration screen.

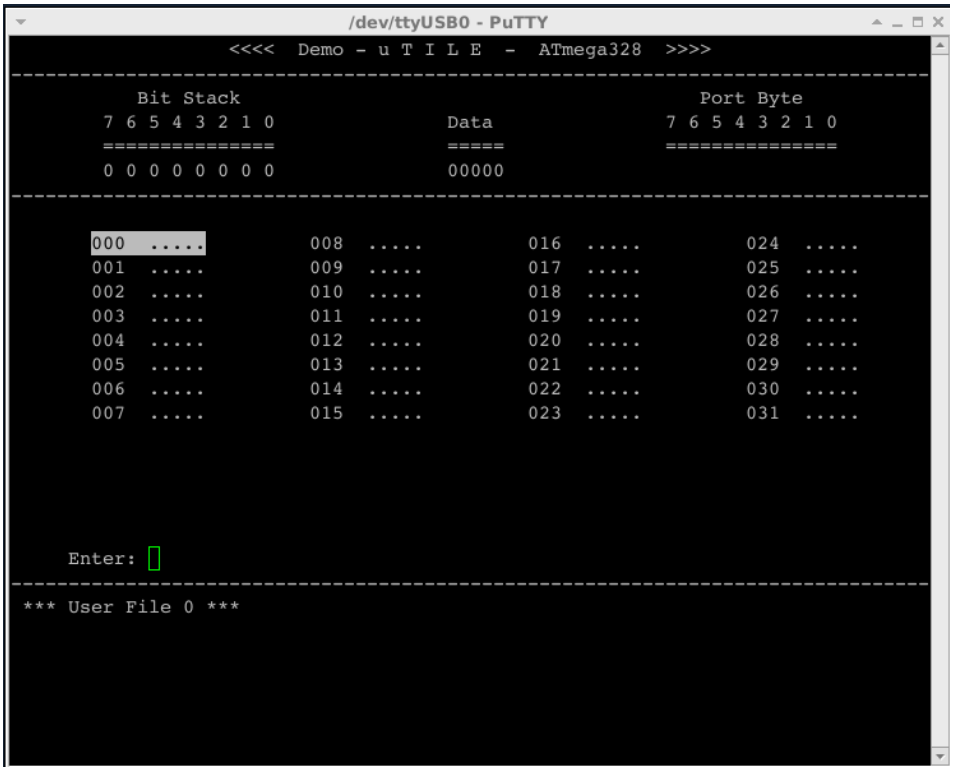
Linux requires a user to belong to the dialout group before access to the USB serial device is granted. Type the command ***groups*** and see if you belong to the dialout group or not. Add yourself to the dialout group with the command issued ***sudo usermod -a -G dialout <username>.*** Change the permissions for ***/dev/ttyUSB0*** with the command ***sudo chmod a+rw /dev/ttyUSB0***, and do the same for ***/dev/ttyACM0*** if you need to access this device for your controller. You only need to do this once.

Start PuTTY and the Configuration screen pops up. Under the Session category, Serial line box, enter /dev/ttyUSB0 (or /dev/ttyACM0 if this is what you saw in the above step). In the Speed box, enter 19200. Then select Serial the Connection type button.

Enter any other configuration settings to suite your preferences. Under the Windows category you could select a window size of 80 columns and 24 rows for a reasonable size screen, and under Fonts used for ordinary text select Courier 10 pitch 14 for larger characters.

You can save the session setup by entering a session name and selecting Save. Next time you start PuTTY you can select your saved session settings to avoid having to re-enter the same information again.

PuTTY should open a terminal screen and uTile main editor screen will appear if the configuration is correct. The screen shot below shows the Demo uTile screen when the controller is powerer up.


/dev/ttyUSB0 - PuTTY
<<<< Demo - u T I L E - ATmega328 >>>>
Bit Stack
7 6 5 4 3 2 1 0
0 0 0 0 0 0 0 0
Data
00000
Port Byte
7 6 5 4 3 2 1 0
000 .....
001 .....
002 .....
003 .....
004 .....
005 .....
006 .....
007 .....
008 .....
009 .....
010 .....
011 .....
012 .....
013 .....
014 .....
015 .....
016 .....
017 .....
018 .....
019 .....
020 .....
021 .....
022 .....
023 .....
024 .....
025 .....
026 .....
027 .....
028 .....
029 .....
030 .....
031 .....
Enter:
*** User File 0 ***

2 Questions and Answers

Q1. PuTTY does not connect to the Nano, why?

A1a. Under Windows make sure you have selected the correct serial port settings and COM port number that is mapped to the USB serial connection used by the Nano. After you establish a successful connection, you should save the setup configuration so PuTTY will continue to use the working configuration when you start it up the next time.

A1b. Under Linux make sure you have selected the correct /dev/ttyxxx0 device that you identified earlier.

Q2. Can I use another terminal emulator if I set the communications parameters to 19,200 baud, 8N1, same as PuTTY, say for example the terminal emulator provided with the Arduino IDE?

A2. You can use any terminal emulator provided it conforms to the VT100 specifications. uTile uses escape control sequences for cursor positioning control and for managing character display attributes. Since Arduino's terminal emulator is not VT100 compatible, it won't work properly with uTile.

Q3. Is there a downside to using PuTTY compared to other communications programs?

A3. Once Putty is configured and working, it provides a fast, lightweight and capable terminal interface with performance on par with the the other terminal emulators. uTile uses only a fraction of the full functionality of the programs, so it is easy on performance requirements.